7 A bar magnet is suspended from the free end of a helical spring, as illustrated in Fig. 7.1.

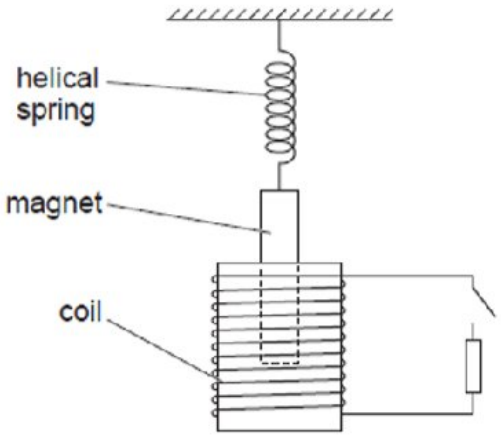


Fig. 7.1

One pole of the magnet is situated in a coil of wire. The coil is connected in series with a switch and a resistor. The switch is open.

The magnet is displaced vertically and then released. As the magnet passes through its rest position, a timer is started. The amplitude of oscillation is constant initially until time t = 3.0 s when the switch is closed.

(a) **(i)** State Faraday's law of electromagnetic induction.

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.....[2]

(ii) State Lenz's law of electromagnetic induction.

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.....[2]

(b) Use the laws of electromagnetic induction to explain why, after t = 3.0 s, the amplitude of vibration of the magnet is seen to decrease.

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.....[4]

[Total: 8]